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Multi-scale model of food drying: Current status and challenges

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Abstract

For a long time, food engineers have been trying to describe the physical phenomena that occur during food processing especially drying. Physics-based theoretical modeling is an important tool for the food engineers to reduce the hurdles of experimentation. Drying of food is a multi-physics phenomenon such as coupled heat and mass transfer. Moreover, food structure is multi-scale in nature, and the microstructural features play a great role in the food processing specially in drying. Previously simple macroscopic model was used to describe the drying phenomena which can give a little description about the smaller scale. The multiscale modeling technique can handle all the phenomena that occur during drying. In this special kind of modeling approach, the single scale models from bigger to smaller scales are interconnected. With the help

of multiscale modeling framework, the transport process associated with drying can be studied on a smaller scale and the resulting information can be transferred to the bigger scale. This article is devoted to discussing the state of the art multi-scale modeling, its prospect and challenges in the field of drying technology. This article has also given some directions to how to overcome the challenges for successful implementation of multi-scale modeling.

1. Introduction

The Important sources of essential dietary nutrients such as vitamins, minerals, and fiber are fruits and vegetables. They are considered, as perishable commodities as the moisture content of them are more than 80% (Paull, 1999). The best way to keep a product fresh is by maintaining its nutritional value. A common technique for a storage system for perishable items is to maintain low temperature so as to preserve the nutritional value, but it is difficult to maintain a low temperature throughout the distribution chain. This contribute to the loss of nutritional value. This is particularly problematic for post harvesting management of fruits and vegetables in developing countries where the distribution chain cannot maintain the required low level of temperatures and thus ruining essential nutritional values of the perishables. An effective answer to this problem is drying. To promote the food security of over 20% of the world's perishable crops, drying is a traditional choice (Sagar & Kumar, 2010). Drying of fruits and vegetables enhance storage stability, minimize packaging requirements and reduce transport weight and cost. (M. Rahman, Mustayen, Mekhilef, & Saidur, 2015). The main purpose of drying is to remove moisture from a material while ensuring the quality of the product. Minimizing operational cost is also considered in the drying process.

Food is a multi-component system which is a complex integration of different chemical and physical process. Foods are mainly composed of proteins and carbohydrate polymers that work as structure builders and the viscosity enricher (Mitchell, 1998). Moreover, there are complex relations between the macroscopic and the microscopic properties of food materials. The explanation of this macro-micro property relation cannot be explained using bulk experimental

methods (Limbach & Kremer, 2006). Mathematical modeling can prove its applicability in the complicated empirical situation (Attig, Binder, Grubmuller, & Kremer, 2004; Baschnagel et al., 2000). A good mathematical model can help to understand the transport process inside the food materials during drying. Drying of Porous materials like food is one of the most complex problems in the engineering field that also has multiscale nature (Mujumdar, 2007). Multiscale simulation method can serve as an effective approach and can extend the vision of macro-micro relationship in food drying process. A multiscale modeling technique can establish the relations between the macroscopic properties with the microstructure of food materials during drying.

The objective of this review is to discuss the application of innovative computational methods especially multiscale modeling approach in the food drying process. This review will also focus on the potentiality and the challenges of multiscale modeling technique. The content of this article proceeds as follows: Firstly, the microstructure of the plant based food materials has been discussed. After that, the water distribution and the transport process in food materials at different scales have been discussed. Next, the article will illustrate the modeling approaches that have been formulated and the limitation of these conventional models. Following that, the multiscale model with, its basics, current state and the solution technique will be explained. After that, the interconnection among the sub-models and the challenges will be expounded. Additionally, several advanced experimental techniques for validating the multiscale model have been discussed. In the final section, some recommendations have been provided for the further development of the modeling technique.

2. Microstructure of plant-based food materials

Plant-based foods are composed of polymers and waters generally. It also contains air and minerals that are responsible for the complex structure (Gross & Kalra, 2002). The complex food structure is built as consequences of multiple interactions of polysaccharides and proteins. Glucose is built by the tissue organization (Street, 1973). Cell structures usually vary widely range in different samples. Cell walls are formed by the integration of cells. The stabilization of the cell wall takes place with the help of fibres. Cellulose, pectin and hemicellulose in the plant cell usually build the cell wall. The water inside the plant cell migrates through the pectin as it is soluble in water (M. U. Joardder, Kumar, & Karim, 2015).

In the case of drying process, the food material is considered as porous and hygroscopic media with multiphase transportation of heat and mass transfer inside it (Ni, Datta, & Torrance, 1999). Fruit tissue, particularly cellular tissue, can be viewed as a summation of parenchyma cells and intercellular spaces (H. Mebatsion et al., 2006). A thin cell wall composed of cellulose and other polysaccharides envelops the cell. The porosity of such cell walls is high, and correspondingly has high permeability to water, so that water retention occurs mainly by surface tension effects. Inside the cell wall and enclosing the cytoplasm there is also a semi-permeable membrane, called the plasma lemma (Jose Miguel Aguilera & Stanley, 1999).

Within each cell, the cytoplasm contains a variety of organelles related to the metabolic activity of the cell, such as the nucleus, mitochondria and others; it is internally bounded by a membrane known as a tonoplast. Inside the tonoplast there is the vacuole, which has the highest water content of the cell phases described, reaching levels of about 98%, together with sugars, organic

acids, salts and colloidal materials. The membranes are semi-permeable and when coming in contact with the vacuolar solution solute concentration, the results in an osmotic effect. The water chemical potential external to the vacuole generates hydrostatic pressure, which maintains the protoplast firmly pressed against the cell wall and hence keeps the tissue firm. This means that loss of water is accompanied by a loss of internal pressure, with the tissue becoming flaccid. This pressure is known as turgor pressure and plays an important role in the rheology and texture of the tissue.

Most of the materials including the food materials contain structures of different scales. These scales are specified by macro-scale, meso scale and micro-scale. For example, when the structure of an apple is analysed different tissues such as vortex, outer, and inner cortex tissues are found. These tissues are considered as elements of the structure. This tissue is formed by the integration of cells that can be observed with the help of a light microscope. Cells have several features *i.e.*, membrane, cell wall that is microscopic in magnitude. On the other hand, if the apple is placed in a dehydration system some physical phenomena can be observed in the centimeter scale which includes moisture loss. This physical phenomenon is closely linked to the cellular features of apples. The discussion above shows that, fruits and vegetables are multi-scale materials. Therefore the drying of food should be explained with the multi scale model for better understanding (Quang T Ho et al., 2013). A demonstration of different structural arrangements is presented in Figure 1.

2.1 Relationship between microstructure and physical property

The drying environment has a great influence on the microstructure. Food structure is constructed of several materials and their mutual interactions (José Miguel Aguilera & Lillford, 2008). The structure is closely related to the quality of the food materials (Ko & Gunasekaran, 2007; Witek et al., 2010). Drying of food includes modification of microstructures, and this can be controlled by controlling drying conditions. Novel approaches for drying process mostly rely on the proper understanding of the architecture and organization of the microstructures.

To design an accurate food processing system, it is imperative to know the structure and the behaviour of the product. Clear knowledge of the relationships between structure and property of the food materials are required for developing a better food drying process (Jose M Aguilera, Stanley, & Baker, 2000). However, the fundamental structure-property relationships are not very clearly understood till now. As a result, food processing engineers rely on the empirical data for designing the drying process. Spectacular advances in materials science during the last decade have emanated from the understanding of the structure of material, its relation to properties (so-called structure–property relationships), and how to engineer and control those properties. Material scientists are quite successful in doing this because the microstructures of metals, ceramics and polymers are more or less homogeneous (Limbach & Kremer, 2006). A large collection of data on physical properties of foods including thermal, rheological, mechanical and colour, exist in literature but understanding their relations to microstructure is minimal (M. U. Joardder et al., 2016a).

It is suggested that the microstructural changes during drying of food materials should be considered as the texture is derived from the microstructure (Bourne, 2002; Mohammad Shafiur Rahman, 2008). In the case of industrial food processing system, the microstructure of the plant-based materials is broken. This happens because of the loss of turgor pressure and rupture of the cellular membranes (Ludikhuyze & Hendrickx, 2001). The microstructure of food material has an auditory perception during biting. The sounds that come out during biting of fresh fruits or vegetable are the instant release of turgor pressure inside the cell (Vickers & Bourne, 1976). In the case of dried product, this sound comes from the breaking of the cell wall (Duizer, 2001).

Different types of drying have a various effect on the microstructure and the product properties. The changes in the microstructure that occurs during drying have an effect on the quality of the food materials (M. U. Joardder, C. Kumar, et al., 2015). The effect of combined hot air drying with the osmotic pre-treatment was evaluated by (Prothon et al., 2001). The microstructural studies showed that the quality of the final product was improved for osmotic pre-treatment. Therefore, a proper understanding of structure-properties relationship will enhance the accuracy of the perfection models related to food processing.

3. Water transport in different scale drying

In fresh fruits and vegetables, water remains inside the cells and some portion of water stays in intercellular space. Researchers have found that 85% to 95% water remains inside the cell which is known as intracellular water (A. Halder, Datta, & Spanswick, 2011). The rest of the water remains in the intercellular spaces. Mainly three transport mechanisms dominate the internal moisture transfer. These mechanisms are as follow: (1) transport of bound water, (2) transport of

free water and (3) transport of vapour. The flow of the liquid water occurs due to the capillary forces. The transport of water vapour occurs by the diffusion and the transport of bound water takes place by desorption and diffusion process (M. U. Joardder, Karim, Kumar, & Brown, 2015).

Water flux inside plant tissue can follow three pathways of transport during drying such as intercellular, wall to wall and cell to cell. The intercellular transport refers to the water vapour movement through the intercellular spaces. The wall to wall transport refers to the capillary water flow through the cell wall tubes. The cell to cell transport refers to water flux through the vacuoles, the cytoplasm and the cell membranes. In high moisture contents, liquid flows due to dominating capillary forces. The amount of liquid in the pores decreases with decreasing moisture content. A gas phase is built up while decreasing the moisture content which causes a decrease in liquid permeability. Gradually the mass transfer is taken over by vapour diffusion in a porous structure with increasing vapour diffusion. However, there are different types of mass transfer mechanisms that dominate at different scales of food materials. Different transport mechanisms of water in different scale are presented in Table 1.

Water transport during hot air drying leads to porosity evolution. Moreover, shrinkage of the food material towards the center of the sample compensates for the void volume. Therefore, the structural deformation phenomenon during heat and mass transfer phenomena is important to understand for designing the drying process. On the other hand, the process parameter and the transfer mechanism influence the structural changes and the transfer mechanism (Rastogi, Raghavarao, Niranjana, & Knorr, 2002; Ziaifar, Achir, Courtois, Trezzani, & Trystram, 2008).

The pore formation is prevented by the shrinkage. The degree of the structural changes depends on the distribution of water within the food tissue (Alamar, Vanstreels, Oey, Moltó, & Nicolai, 2008). Water can be classified on its spatial basis, as intercellular water and intracellular water. Intracellular water remains inside the cells while the intercellular water remains in the capillaries. Water in the intercellular space water varies between 78-97% (wet basis) in different fruits and vegetables. Considering the structure-properties relationship and water migration mechanisms, there are several ways of prediction of drying kinetics and deformation in the literature. The following sections describe the existing challenges associated with those models.

4. Modeling of Drying

For discovering the possible solution to a given problem in food drying, mathematical modeling is an efficient tool. The classification of mathematical modeling of food drying can be done by two basic scales. One is macro-scale model, and another is a micro-scale model. The combination of these two models is named as a multiscale model. Macro-scale models can be categorized in two approaches such as theoretical model and empirical model. The empirical models are formulated based on the experimental data, and there is no physics associated with it (A. Halder et al., 2011). These models are simple and easy to apply for developing drying kinetics. Theoretical models, on the other hand, are based on physics associated with heat and mass transfer during drying. Developing physics-based models are challenging tasks. The complex nature of the agricultural product and the changes in the thermo-physical properties during drying are the main hurdles to formulate the physics-based models. Moreover, the heat and mass transfer phenomena are highly coupled in the drying process. The micro-scale model is

a comparatively new approach. It is a very important approach towards multi-scale modeling. The classifications for the mathematical modeling of food drying have been presented in Figure 2.

4.1 Empirical models for food drying

Empirical models are easier to formulate as the mathematical computation is simple. The output parameters are measured by the data logging system attached with the experimental setup. A block diagram for the empirical modeling approach is presented in Figure 3.

Numerous empirical models have been formulated for different food stuffs. The most popular models are Page, modified Page models, Newtown's and Lewis models (Erbay & Icier, 2010). The drying curve of each experiment has been plotted by measuring the moisture content at different drying time. The basic exponential model is presented as follows

$$MR = \frac{M - M_e}{M_0 - M_e} \quad (1)$$

The basic empirical model got wider acceptance among the researchers as these models can describe the drying behavior but these models failed to describe the complex phenomena of the drying process. Moreover, these drying models consider moisture content as a function of drying time and fail to describe the temperature and the moisture distribution in the sample during drying. It is very crucial to study the moisture content through the food materials for analyzing micro-organism activity inside it. The micro-organism activity is very important for the food quality and preservation process. For this reason, a robust mathematical modeling is required that

can explain the simultaneous heat and mass transfer phenomena in a variation of moisture content and temperature.

4.2 Theoretical modeling

Drying is a multi-physics problem where coupled heat and mass transport processes occur inside the material simultaneously. Therefore, theoretical modeling can provide a realistic understanding of drying kinetics that can help for optimizing energy efficiency and quality attributes. Theoretical modeling can be categorised into two types: single phase modeling and multiphase modeling. There are many single phase models of food drying (Batista, da Rosa, & Pinto, 2007; Kaya, Aydın, & Dincer, 2008; Kumar, Millar, & Karim, 2015; Paull, 1999; M. Rahman et al., 2015; Sagar & Kumar, 2010) that have been developed based on the concept of bulk moisture transport. Diffusion is taken into consideration as the mechanism of the moisture migration in single phase model. The basic mathematical equations that have been considered for developing single phase model are shown in Table 2.

Single phase model only considers conduction of energy transport and diffusion for mass transfer. These models have many significant limitations, for instance, the models have totally lumped for heat and mass transport and do not include many important physics such as pressure-driven flow or transport due to physicochemical changes in the porous medium.

In addition, due to the hygroscopic, porous and amorphous nature, plant-based food materials contain three phases such as solid, liquid water, and gases including water vapour and air (Curcio, 2010). Therefore, multiphase porous media model for simultaneous heat and mass transfer can provide a realistic understanding of transport processes. Multiphase model for food

drying is the model where three phases that (liquid water, gasses and solid) have been considered for coupled heat and mass transfer. Therefore the development of the physics-based multiphase model is a challenging task because plant-based food material is non-homogeneous and hygroscopic in nature. There have been some attempts by the researchers to develop a multiphase model for different food processing such as frying (B. Farkas, R. Singh, & T. Rumsey, 1996; A. Halder, A. Dhall, & A. Datta, 2007a; H. Ni & A. Datta, 1999b; A. Warning, Dhall, Mitrea, & Datta, 2012; R. Yamsaengsung & R. Moreira, 2002), heating (A. Halder & Datta, 2012; J. Zhang & Datta, 2004), baking (Feyissa, Gernaey, Ashokkumar, & Adler-Nissen, 2011; H. Ni & A. Datta, 1999a; Abdellah Ousegui, C Moresoli, M Dostie, & B Marcos, 2010; Papasidero, Manenti, & Pierucci, 2015), cooking (Dhall, Halder, & Datta, 2012; Van der Sman, 2013) and drying (Datta, 2007a; Gulati & Datta, 2015; Kumar, Joardder, Karim, Millar, & Amin, 2014; S Mercier, B Marcos, C Moresoli, M Mondor, & S Villeneuve, 2014). The researchers developed their multiphase model using the basic mathematical equations that are given in Table 3.

Furthermore, different researchers developed multiphase model considering two different approaches namely, sharp boundary model and distributed evaporation model. These types of formulation are sometimes called moving boundary, interface tracking or front-tracking problem. On the other hand, the evaporation that is considered to be distributed throughout the sample domain is treated as distributed evaporation (A. Halder, Dhall, & Datta, 2010; Kumar C, 2015). Most of the multiphase model has been developed considering sharp boundary evaporation in different food processing system, for drying (Curcio, 2010; Khan, Kumar, Joardder, & Karim, 2016; S. Mercier, B. Marcos, C. Moresoli, M. Mondor, & S. Villeneuve, 2014), for frying

(Bouchon & Pyle, 2005; Farid & Chen, 1998; B. E. Farkas, R. P. Singh, & T. R. Rumsey, 1996) and for baking (A. Ousegui, C. Moresoli, M. Dostie, & B. Marcos, 2010). However, the problem of the sharp-boundary formulation is that it can predict the phase change occurs at the dried (crust) and moisture material (core) interface only. That is why distributed evaporation model is necessary which can predict a narrow evaporation zone where rapid evaporation is taking place. The sharp-boundary model has other limitations such as over prediction of the core temperature as it ignores any phase change in the core that would act as a heat sink. Since transport of water vapor toward the core is ignored in a sharp boundary formulation, migration of water vapor from the interface to the core does not occur during these types of formulations. Sharp boundary problem formulation also over-predict the vapor flux in the crust region as the pressure difference between the surface and the interface is never as high as 1 atmosphere (A. Halder, A. Dhall, & A. K. Datta, 2007).

Moreover, based on the thermal properties, the multiphase model can be categorized as equilibrium and non-equilibrium formulations. Most of the authors have developed their models considering equilibrium conditions (Curcio, 2010; S. Mercier et al., 2014; H. Ni & A. K. Datta, 1999; R. Yamsaengsung & R. G. Moreira, 2002). Equilibrium conditions, though have some problems. Transport will affect the establishment of equilibrium condition and it may be valid for high moisture content materials. For food processing; one needs to be mindful that equilibrium condition may never be achieved at the surface since transport is strong there. Therefore, the non-equilibrium formulation is needed for predicting accurate heat and mass transfer. Some researcher developed their model by considering non-equilibrium formulation (A. D. Halder, A. Datta, A. K., 2007; 2012). However, they considered food as a non-hygroscopic

porous material and ignored the physics of bound water transport in their model. In practice, plant-based food materials are hygroscopic in nature, and it contains three types of water that are discussed below.

The above-mentioned water transport mechanism should be included during developing heat and mass transfer model for food drying. However, in multiphase models mentioned above have not considered the bound water transport through these mechanisms during their modeling. As the multiphase models are considered as a macroscopic model, therefore, it is very difficult to apply the boundary conditions of cellular level water transport in macroscopic level. This is a major limitation of existing multiphase model. Though, some researchers have attempted to develop multiphase model considering bound water transport (A. Halder, A. Dhall, & A. Datta, 2007b; H. Ni & A. Datta, 1999a), they eventually developed the model for wood drying. (Feng, Tang, Cavalieri, & Plumb, 2001) Developed a multiphase model considering bound water transport for porous material. They claimed that the model was developed for food drying. However, they used the mass conservation equation and the property for wood drying. Major limitations of the above-discussed models were that they did not consider material shrinkage for migration of bound water.

Migration of bound water has a significant effect on the material structure. Migration of intracellular water causes cell shrinkage, pore shrinkage as well as the collapse of cell structure whereas the movement of cell wall water causes overall tissue shrinkage (Rakesh & Datta, 2011). Therefore, for obtaining micro-level observation of different physics and prediction of transfer mechanism and deformation macroscale multiphase model is not a proper option.

Considering the above discussions, multiscale modeling of coupled heat and mass transfer during drying would be the best approach regarding energy efficiency as well as food quality.

4.3 Challenges in existing modeling technique

The effect of drying conditions on the microstructure of food materials has been neglected for a long time. For estimating the molecular diffusion and explaining the mass transfer mechanism Fick's first and second law are applied without knowing the actual mechanism of the mass transfer (José Miguel Aguilera, 2005; H. Mebatsion et al., 2006). Nevertheless, developing the mathematical models and calculating mass transfer rate remains incomplete unless the microstructure is resolved. The microstructural elements that are below 100 μm range critically participate in transport properties and the physical behaviour of food material (Jose M Aguilera et al., 2000). Moisture transport through the pores, cell and cell walls of food materials occurs at this micro-scale. In this respect, microscopic feature is required for modelling to explain the heat and mass transfer process at this microstructural level. Most of the plant based food materials encompass parenchyma tissue. The parenchyma tissues are randomly distributed arrays of pores and cells (Fanta et al., 2012). It has been claimed that the microstructure can be generated with the help of microscopic image taken by the Atomic Force Microscopy (AFM) with proper image processing algorithm (H. Mebatsion et al., 2006).

Plant-based food materials are assumed to be linearly elastic on the macroscopic scale. Based on this assumption, the predictive models for food drying have been formulated (J. Zhang, Datta, & Mukherjee, 2005). The microscopic feature also has a good effect on the macroscopic properties of the food materials. The geometric properties of the cell, cellular water potential, mechanical

properties of the cell wall, the presence of fluid in the cell as well as inter cellular spaces are good features of microstructural properties (Van Liedekerke et al., 2010). As these features are useful, microstructural properties should be included for the modeling of the drying of food materials. The effect of cell geometry on the physical properties of the materials is not understood comprehensively to date (Juge, 2006; Lee et al., 2004; McCann et al., 2001). Within this context, the food drying models which have been developed so far face difficulties in validation process (H. K. Mebatsion, Verboven, Ho, Verlinden, & Nicolaï, 2008). In plant-based food materials, the variability of components can be found in all scales. As there is a complexity in the spatial arrangement of the food architecture microstructural geometry of food is required. In this respect, multiscale modeling of drying of food materials is very important for food drying technology.

5. Multiscale modeling technique

For a better understanding of water transport and the mechanical behaviour during drying of food materials modeling are an effective tool. There are two basic approaches to model the drying phenomena of food materials. One is macroscopic continuum approach, and another is the microscopic (cellular) approach. In the first approach, the tissue is considered as homogeneous, and the modeling is carried out based on the lumped properties of the cell wall and pores (Nguyen, Dresselaers, et al., 2006; Veraverbeke, Verboven, Van Oostveldt, & Nicolaï, 2003). The other is micro-scale modelling approach where, the tissue is considered as heterogeneous and the complex cellular structure is represented by the geometric model. Most of the works regarding the food materials drying are based on classical continuum approach where the

transport properties are considered as a lumped single value throughout the tissue. The tissue of food materials undergoes large deformations during drying (Karunasena, Gu, Brown, & Senadeera, 2015). The cell wall deformation occurs as a consequence of turgor pressure reduction acting on the cell wall. The macroscopic properties and the tissue strength of food materials are greatly influenced by the cell size and shape, turgor pressure as well as the presence of intercellular space (Konstankiewicz & Zdunek, 2001). For better understanding of the influence of drying condition on the tissue the micro-scale model is an effective approach. The main drawback of the macro-scale model is that, it does not provide the platform to investigate the microstructural behaviours of the tissue during drying which can be overcome by the micro-scale modelling. The micro-scale model can describe the deformation of cell wall and the local distribution of water in cellular level. For predicting, the drying process of entire fruit, the microscale model can be extended. The extended micro scale can be expressed as multiscale model. The multiscale model can provide the detail model of microstructural features combining the continuum type models. The concept of multi-scale modelling addresses some of these limitations, such as the absence of equilibrium at the microscopic scale or the consideration of product variability at the industrial scale. The concept of multiscale modelling in drying technology is presented in Figure 4.

5.1 Present status of multiscale modeling

To establish the relationship between the pore scale mechanism and the macro-scale phenomena multiscale model can play a key role in drying technology. The multiscale modeling approach is relatively new in the food engineering sector, and only a few representative publications are

available. Some of the significant works related to the multiscale modeling are presented in Table 4. The table gives an overview of advance modeling approach for mostly related to drying technology.

Nicoli et al. applied the multi-scale modeling to the post-harvest storage of vegetable and fruits (Nicolai, Verboven, Scheerlinck, & De Baerdemaeker, 1998). To compute water diffusion coefficient in apple skin micro-scale models by (Veraverbeke et al., 2003). For describing the metabolic gas exchange in pear fruit during post-harvest storage, a multi-scale model was developed by the researchers (Q Tri Ho, Verboven, Verlinden, & Nicolaï, 2010). The micro-scale in the intercellular spaces through the cell wall was obtained from the radiation tomography image. The cellular respiration model was also developed (Quang Tri Ho et al., 2009).

A multi-scale approach to heat and mass transfer in porous media with wood drying was formulated by Patrick Perre (Perré, 2010). Some simulation of multi-scale transfer configuration was proposed considering the heterogeneous porous medium. A stack of timber was dried in the convective dryer. For considering several scales at the same time, homogenization process was applied. A concept of multi-scale presented is presented in Figure 5.

The principal concern of (Perré, 2010) was the pore morphology, and a couple of computer codes were formulated to present the modeling works. The computer codes are named as *Mesh pore* and *Morpho pore*. These computer codes were formulated based on the image-based meshing and the homogenization process. Later a new dual scale model was presented where the researchers were trying to initiate a new approach to multi-scale modeling without the classical homogenization complexity (Carr, Turner, & Perre, 2013). However, only 2D analyses were

done in these models. Therefore, it is still not clear about the 3D phenomena that occur during drying. Moreover, the coupled heat and mass transfer phenomena were considered for wood drying. In the case of food drying, quality is also a very important parameter (Nijhuis et al., 1998). If this model is applied to the food drying process, the consequences of the fluid transport and the effect of cell wall properties on the shrinkage phenomena will not be explained well.

A particle-based approach for simulating the micro-level was designed by the researcher (Karunasena, Senadeera, Brown, & Gu, 2014). The influence of the change of moisture content on the cell wall and the cell area was investigated in this work. The cell wall was considered as the non-linear elastic material. Though it is a great approach in this field which gives a lot of information related to the micro-level drying, it has some limitations. The cell and the tissue structure of the food materials do not follow a regular shape (Abera et al., 2013). It is really hard to explain the real phenomena of large deformation of the cell and tissue with approximated structure. Therefore, a new approach is required to illustrate the real microstructure of food materials and explain the large deformation in cellular level drying.

When the microscopic scale is much smaller than the macroscopic one, the coupling between scales is sequential, and the multi-scale approach reduces to a change of scale, also known as scaling (Defraeye, 2014). On the contrary, when a local thermodynamic equilibrium is not attained, the scales usually overlap, and two-way coupling is required. This is a real multi-scale configuration, more demanding both in computational resources and applied mathematics. Drying has a distinct multi-scale nature on exchange and the transport process at different scale. One of the strategies for the interconnection is homogenization.

The biophysical activity in a cellular level depends on the gas transportation in the intercellular territory. For explaining the vapor and the fluid transportation during drying of food materials modeling is the best way. Some researchers have initiated serially coupled multi-scale model for avoiding the homogenization process (Esveld, Van Der Sman, Van Dalen, Van Duynhoven, & Meinders, 2012; Esveld, van der Sman, Witek, et al., 2012). The paradigm can predict the dynamic moisture diffusion into cellular solid food. The air pores and the air connectivity of the cell were extracted by X-ray tomographic image and made a discrete microstructure model. A multi-scale model of the mechanical properties for the food baking process was formulated (Guessasma, Chaunier, Della Valle, & Lourdin, 2011).

Numerous efforts have been made to model the spatial distribution of gas and the moisture exchange, but the mathematical models remain incomplete as they did not consider the microstructure (Putranto & Chen, 2015; A. D. Warning, Arquiza, & Datta, 2015). As a result, a new model is required for describing the actual phenomena at the cellular level in the case of food drying.

5.2 Microscale geometrical model

In the microscale model, a different component of cells is distinguished regarding the position, shape, and orientation. The microscale features of the component such as cell, cell wall and cell membrane are also studied. The effect of microstructure on transport phenomena in food drying process has been neglected for a long time while developing the transport models. But the transport phenomena inside the food materials, physical and biochemical stability greatly depend on the microstructure. In the case of drying technology, Fick's first and second laws are used to

estimate the rate of mass transfer. The mass transfer investigation remains incomplete unless the structure is resolved. It is evident that the cell wall can be a controlling factor in the mass transfer process. Extensive approximation has been made by the food engineers for the solution of the diffusion equation in a regular geometry. Moreover, the microstructure of food materials is not regular, and there is no model to investigate the effect of drying process in food microstructure.

A representative volume element is defined as the minimum volume over which the properties of the material can be calculated. The RVE has been chosen as a non-hygroscopic material where the solid matrix is considered as the cell surrounded by free water inside it. A schematic diagram of the RVE is presented in Figure 6.

In practice, plant-based tissue is hygroscopic in nature in which the cells are filled with fluid commonly known as bound water. Therefore, it is necessary to choose a perfect Representative Elementary Volume (REV) to develop a model for better prediction. The schematic diagram of the microstructure of cellular tissue is presented in Figure 7.

The actual microstructural geometry for developing a better model is presented in Figure 8. This geometry will work as a computational domain which is a compulsory requirement for developing a model.

5.3 Modelling microstructures of food materials

The microstructural information generated can be used to understand transport mechanisms during drying and to assess the functionality of finished products. Efforts should be made to

derive appropriate scaling laws for results obtained at the microstructural and nano-structural levels. A variety of microscopic techniques is available for studying different structures and components of food systems. Building the micro structure of the biological system is the most complex system in the process of formulating the multi-scale modeling. Despite a great deal of momentum in cell biology and fruit science, microstructural characterization and computational modeling of biological systems is still in its infancy. It is not possible to analyze the heat and mass transfer by means of finite element method due to lack of proper microstructure. Scanning Electron Microscope (SEM) imaging provides detailed microstructure information at micro scale resolution without chemical treatment. A suitable image processing technique such as Voronoi tessellation algorithm can extract the geometrical data and provide a clear idea about the microstructure. A schematic block diagram of the development microstructure from the microscopic image is presented in Figure 9.

The transport of water in intercellular spaces and, the cell walls network is predicted by using the chemical potential as the driving force for water exchange between different microstructural compartments. The transport property of the water in the cell wall network can be obtained from the literature.

6. Numerical Solution for the models

Several advanced numerical methods for the solution of microscale as well as multiscale modeling have been used. This section will discuss the numerical solution techniques that have been used for solving the Multi-physics related problem at various scale.

6.1 SPH and DEM method

The key idea in so-called mesh less particle methods is that the material is mass-discretised into material points. Some researchers have been working on particle-based numerical model to simulate micromechanics of plant cell (Karunasena, Senadeera, Brown, et al., 2014). Some of them are concerned with addressing how bulk tissue level mechanics are related to the micromechanics of cells. They have developed mesh free particle methods to simulate the mechanics of both individual plant cells and its response to the external stress. At the same time, there has been a significant concern on modeling the mechanical behavior of tissue. The model considered the cell fluid and the cell wall as the two main features of a plant cell. For developing the cell fluid model, Smoothed Particle Hydrodynamics (SPH) has been used. On the other hand, Discrete Element Method (DEM) has been used to model the cell wall. In this model, the researcher has tried to address how biological cell responds to mechanical load. A two-dimensional single cell model that is capable of predicting cellular level shrinkage as a function of moisture content has been developed. The researchers claimed that the mesh free nature of the model gives it the capability to handle large deformations and multiphase interactions and subcellular details which are the key concern in multiscale modeling approach (Karunasena, Senadeera, Brown, et al., 2014).

6.2 Finite element method

For solving the partial differential equation, finite element method (FEM) is a flexible and widely used method (Zienkiewicz & Taylor, 2005). In FEM method the computational domain is subdivided into a number of elements having different shape and size. Every element in the

computational domain are interconnected in a certain number of nodal points. An unknown solution is assigned to every element. This unknown solution is known as shape function. In the following step, the governing equations are discretised using suitable meshing technique. The governing equation is normalized with this meshing technique. Depending on the time variable the system equations are defined whether it will be ordinary differential equation function or algebraic equations. After that, these equations are solved by the finite difference approximation approach. The details of finite element method can be found in many literatures (Bathe, 2008; Efendiev & Hou, 2009).

6.3 Lattice Boltzmann method

To simulate of the microscale and the meso-scale model the lattice Boltzmann method and the molecular dynamics methods are also useful. Lattice Boltzmann method is different from the traditional approaches (Van der Sman, 2007). Materials and the fluids are represented as quasi-particle in the lattice Boltzmann method. These particles maintain the basic conservation law of energy, momentum, and mass while interacting via collision. The rules of the collisions maintain the discretised Boltzmann equation. The discretization of momentum, space and time make Lattice Boltzmann different from the other methods.

6.4 Molecular dynamics

In the molecular dynamics method, the momentum of the molecules is computed with the solution of the Newton's equation of motions (Van der Sman, 2007). The van der Waals, the covalent bond and the electrostatic interaction creates a potential field. The created potential field is used to compute the forces between the molecules. The evaluation of these potentials is

computationally the most intensive step of a molecular dynamics simulation. Molecular dynamics can be considered as a discrete element method. This method is not used frequently in food processing engineering (Limbach & Kremer, 2006). Especially in drying technology this methodology has not been used yet.

6.5 Bridging between the scales

A multiscale model is a hierarchy of some sub-models that describes the process phenomena at different spatial scale (Aregawi, Abera, Fanta, Verboven, & Nicolai, 2014). The main challenge of formulating the multiscale model is to establish a logical relation between the scales (Perré & Rémond, 2006; Perré, Rémond, Colin, Mougel, & Almeida, 2012). Drying of food materials is a multiscale phenomenon and at a better approach to formulating the drying model is the multiscale model (Quang T Ho et al., 2013). For bridging the macro-scale and the microscale models, an effective approach was proposed which is known as “homogenization”. It is a powerful mathematical tool, and the macroscopic properties can be predicted from the microscopic description by using this tool. The homogenization process includes a couple of second order diffusion equations. These equations are known as the governing equation. The apparent diffusion can be calculated by solving the cellular diffusion equation. A sample homogenization approach is presented in Figure 10. (Q Tri Ho et al., 2010).

Recently, Van Der Sman et al. (2012) have developed a multiscale model with serial coupling technique (Esveld, Van Der Sman, Van Dalen, et al., 2012; Esveld, van der Sman, Witek, et al., 2012). The model can predict the dynamics of the moisture diffusion of food materials in cellular level. To develop the model the researcher used the X-ray micro tomography image to get the

information about the cellular network and the pores. The model considers the local diffusivity through the moisture sorption and the pores. These parameters were incorporated into the macro-scale model for predicting the dynamic moisture profile of crackers.

Although researchers have always shown more interest in the macroscopic scale, the microscopic transport phenomenon is also an important factor in drying technology (José Miguel Aguilera, 2005; Fanta et al., 2014). For simulation of the drying process, numerous mathematical equations have been proposed by (Whitaker, 1977, 1998). Later some researchers have made some changes in these equations and assumption and proposed some new models (H. Ni & A. Datta, 1999a). Some of the researchers have considered internal over pressure and the bound water diffusion (Hills & Remigereau, 1997). Some researchers have considered the evolution of the porosity in their model (Datta, 2007a; Ni et al., 1999). Usually, the microscale model is formulated to calculate the apparent properties of the sample more accurately. The Heterogeneous Multiscale Method (HMM) can be a suitable option for coupling both the scales as the constitutive relation relies on many variables (E et al., 2007). The HMM can be initialised by discretization of macro-scale equations. A finite element method is an efficient option in this regard. The framework for the heterogeneous multiscale modeling is presented in Figure 10.

The general framework is as follows. If a microscopic system is given, the state variable of which is denoted by u , and can be written as Equation 2,

$$f(u, b) = 0 \quad (2)$$

Where b is the set of auxiliary conditions, such as initial and boundary conditions for the problem. Here, the microscopic details of u are not interest rather macroscopic state of the system which is represented by U that satisfies the following macroscopic equation:

$$F(U, D) = 0 \quad (3)$$

Where D represents the macroscopic data that are necessary for the model to be completed. Considering the compression operator Q that maps u and U , and R any operator that reconstructs u from U we can find that,

$$Qu = U \quad RU = u \quad (4)$$

Where, compression and reconstruction operators are similar to the projection and prolongation operators used in Multi-grid methods. The goal of HMM is to compute U using the abstract form of F and the microscopic model. It consists of two main components; one is selection of a macroscopic solver another is estimating the missing macro-scale data D using the microscale model.

The microscale model can be used to compute the deformation and the dehydration of tissue of plant-based food materials during drying. The apparent properties of the food materials can be calculated from the cellular properties. For example, the apparent diffusion coefficient can be determined from the apparent water conductivity by following equation (Aregawi et al.,2014)

$$D_{app} = \frac{K_{app}}{\rho_m c_w} \quad (5)$$

Where, c_w is the water capacity of the cell, ρ_m is the density of the cell

Similarly. The apparent water conductivity is calculated from the following relationship (Aregawi et al.,2014)

$$K_{app} = -J \frac{h}{\Delta\phi} \quad (6)$$

Where, J is the total flux of the fruit tissue ($\text{kg m}^{-2}\text{s}^{-1}$), $\Delta\phi$ is the water potential difference in the two side of the cell wall (Pa).

The apparent deformation, which is known as shrinkage of the food materials can be calculated by assuming the change in displacement in the tissue due to a change in turgor pressure. The turgor pressure generates stress that leads to expansion of the cell wall of the plant based food materials. Therefore, the turgor loss can be responsible for the shrinkage phenomena

7. Required properties for the multiscale model

Material properties are the most important part of theoretical modeling. It is required for all the relevant physics for realistic prediction of drying mechanism. The food drying related properties are rheological, mechanical, thermal, mass transport and electrical properties. Food materials undergo changes in moisture, temperature, and structure. Major changes in structure can easily be seen by naked eye. The changes in structure mostly depend on the variation in temperature and moisture. It also varies in different products. So properties of the food materials are required to be known for a better modeling approach. Nevertheless, some properties of food materials are not available in the literature; especially, the cell-level properties of food materials are very rare.

Modeling can contribute to the process design in a significant way, and it can start with the limited available data. If there is no available data in the literature, modeling can start with approximate data with simplified correlation as a starting point. After that, the sensitivity analysis on the unavailable data needs to be completed. Some of the properties are discussed in the following sections.

7.1 Cell shrinkage

Modeling of shrinkage during drying of plant-based food material is an important task. From the mechanical point of view, cells can be considered as closed thin cell wall structure. To model the shrinkage mechanism, Newton's second law was considered by researchers. The following set of equations were solved for the position of the vertices and the velocity of cell wall during drying

$$m_i \frac{dv_i}{dt} = F_{T,i} \quad (7)$$

$$\frac{dx_i}{dt} = v_i \quad (8)$$

Where m is the mass of the vertex (Kg) of the cellular tissue. To simplify the model, the rate of change of the vertices is considered as equal to the net force acting on the vertex.

The cell wall tension is maintained by the turgor pressure and the cell shrinkage is the result of the loss of turgor pressure on the cell wall. The walls of a cell and the adjacent cell walls have been modelled as parallel elements. The cell walls are also considered as non-linear elastic materials. Neo-Hookean elastic model can be used for modeling the cell wall. The total force acting on a cell can be expressed using the following formula

$$F_T = F_w + F_d \quad (9)$$

The force F_w is the resultant of the net turgor pressure. F_d Is the damping force and it is a product of damping factor and vertex velocity. The details of cell shrinkage can be found in (Fanta et al, 2012).

7.2 Cell mechanics

Linear elasticity can describe the small deformation. The mechanical equilibrium without external force can be expressed by the following equation

$$\Delta \cdot \sigma = \nabla \cdot D(\varepsilon^e) = 0 \quad (10)$$

Where D is the stiffness matrix of linear elastic model (Pa) and ε^e is the elastic strain

The shrinkage strain and the mechanical strain formulate the total strain

$$\varepsilon = \varepsilon^e + \varepsilon^d \quad (11)$$

Where, ε^d is the shrinkage strain

As food materials undergo large deformation during drying, it is not possible to quantify the mechanical behavior with the linear elastic model. The nonlinear characteristics of food materials can be explained by the nonlinear elastic model. The stress inside the tissue can be calculated from the following mechanical equilibrium, ignoring any surface and body forces.

$$\nabla \cdot (S \cdot F^T) = 0 \quad (12)$$

Where, F^T is the transpose of the deformation gradient tensor and S is the second Piola-kirchhoff stress tensor. Equation (12) can be used for building the relationship between shrinkage strain and elastic strain.

The nominal stress can be expressed as follows

$$p = \frac{\partial W}{\partial \lambda} \quad (13)$$

Where W is the strain energy density function, and λ is the Stretch ration.

The Neo-Hookean model for the strain energy density function is as follows:

$$W(I) = A_{10} (I_1 - \lambda_1) \quad (14)$$

Where, I_1 is the invariant of the coordinate system, A_{10} is the co-efficient of the energy density function, and λ_1 is the Lames constant also known as stretch ratio.

Lame's constant (λ_1, λ_2)

$$\lambda_1 = \lambda, \lambda_2 = \sqrt{\lambda} \quad (15)$$

$$I_1 = \lambda_1^2 + \lambda_2^2 \quad (16)$$

Therefore,

The nominal stress according to Neo-Hookean model,

$$p = 2A_0 \left(\lambda - \frac{1}{\lambda^2} \right) \quad (17)$$

The value of A_0 can be found from the nominal stress-strain by curve fitting method.

7.3 Cell diffusivity

Cell level diffusivity is an important parameter while considering the cellular level drying. Water diffusion through cell wall can be expressed by Fick's first law considering water moves as a result of pressure gradient (Nguyen, Verboven, Scheerlinck, Vandewalle, & Nicolai, 2006; Nobel, 1999). It can be expressed as follows

$$j = -k \nabla \varphi \quad (18)$$

Where j is the water flux ($\text{kg m}^{-2} \text{s}^{-1}$), φ is the pressure (Pa), ∇ is the gradient operator and K is water conductivity. The relationship between bulk diffusivity and the water conductivity of the cell wall can be derived by Fick's second law of diffusion, as shown below:

$$\rho_{dm} C_\varphi \frac{d\varphi}{dt} = \nabla K \cdot \nabla \varphi \quad (19)$$

Where, ρ_{dm} is the density of the dry matter in cell (kg m^{-3}), C_φ water capacity of the cell ($\text{kg kg}^{-1} \text{pa}^{-1}$) and t is the time (s). The values of water capacity can be found from the sorption isotherm curve. By using constant isotherm properties, diffusivity of the cell can be derived by the following relationship:

$$D = \frac{K}{\rho_{dm} C_\varphi} \quad (20)$$

Where, D is the cell diffusivity.

The researchers have considered the mass balance equations for modeling the natural dehydration of plant-based food materials as these processes are mainly dominated by the mass transfer process. Therefore, heat transfer process is generally ignored in these models. However, drying process involves heat transfer along with the mass transfer process. Therefore, the heat and mass balance must be coupled during modeling the food drying process (Perré, 2015). Multiscale modeling is a comparatively new approach in the drying technology. In the modeling of microscale drying process, no researchers have considered the coupled heat and mass transfer process. The future researchers can focus in this area.

7.4 Capillary diffusivity

Capillary diffusivity is caused due to moisture gradient and temperature gradient. When the material is wet, the effective diffusivity is close to the capillary diffusivity (Datta, 2007b). The relationship between moisture content and the capillary diffusivity in potato is given as follows

$$D_w = 1.0 \times 10^{-8} \exp(-2.8 + 2.0M) \quad (21)$$

Where, D_w is the capillary diffusivity and M is the moisture content of the materials.

7.5 Porosity

Porosity in food material is known as the volume fraction of air space or void space inside the tissue. It depends on the moisture content and the drying condition (Madiouli, Sghaier, Lecomte, & Sammouda, 2012). The void space is termed as pores, and it can be divided into three groups

such as open pores, closed pores, and blind pores. In the open pores, the flow of fluid typically takes place. The blind pores have one close end, and the closed pores are closed from all sides (Datta, 2007b). Porosity in non-hygroscopic materials is not variable with the moisture content. On the other hand, porosity in hygroscopic materials increases with the reduction of moisture content (Marousis & Saravacos, 1990). The gas phase is present in open and closed pores. The open pores are interconnected with the external surface of the materials. The measurement of the porosity can be calculated from the values of theoretical and practical data by measuring the bulk density and the particle density.

For plant-based tissue, porosity exists at their initial condition, and that type of porosity is identified as initial porosity. For example, the initial porosity of apple tissue is around 0.25-0.3 depending on varieties (M. Joardder, Brown, Kumar, & Karim). Porosity can be measured experimentally in many ways, and interested readers can be referred to the works of Joardder et al (2015) for more details. On the other hand, there are many empirical and theoretical model of porosity prediction (Krokida, Zogzas, & Maroulis, 1997; M. S. Rahman, 2001; Ross, Campanella, & Okos, 2002) available in the literature. One of the suitable models of porosity can be used in the heat and mass transfer equation of the multiscale modeling to represent porosity.

8. Model validation

For validating the multiscale modeling approach detailed experimental data is required. The drying of food material is a complex multi-physics and multiscale problem. The complexity occurs because of the non-uniformity, anisotropy and the heterogeneity of the food materials. It needs modern experimental techniques and the development of new drying equipment for

monitoring active drying phenomena. A schematic diagram of the modern laboratory drying equipment is presented in Figure 12.

Experimental techniques based on non-destructive imaging show promising perspectives for detailed validation of numerical models. Most popular ones are presented in Table 5 together with a few applications.

A more in-depth discussion can be found in the literature (Perré, 2011). There is a strong relation between the attainable spatial resolution in the image and the image acquisition time, which both depend on the used techniques and the types of devices used e.g. laboratory, synchrotron or clinical X-ray. The high spatial resolution of some techniques allows studying transport processes down to the micro scale level.

8.1 Microscopic imaging

For obtaining the spatial fields, several microscopic imaging devices are usually used such as scanning electron microscope, transmission electron microscope, optical microscope, confocal microscope, CCD cameras. A small change in the imaging environment can be made during or before imaging to increase the focus on specific information (Paredes, Martinez-Alonso, & Tascón, 2003). Various cellular level properties of the food materials such cell density, moisture content, rheological behaviour, chemical composition can be obtained depending on the method. By combining several images of different angle multi-dimensional, information can be obtained (Garini, Vermolen, & Young, 2005). All of these data are important to understanding the mechanism of food drying in cellular level.

Scanning electron microscope, atomic force microscope and scanning tunnel microscope are the popular device that produces two dimensional images by scanning sample. The scanning process is very slow therefore these techniques are only used when high resolution information as an example up to 1 micrometre is needed (Centonze & Pawley, 2006). Optical microscope, Coupled Charge device (CCD) cameras, Infrared (IR) cameras provide high resolution images in a non-invasive way. IR cameras are suitable for obtaining the temperature profile of the sample (Maldague, 2002). Another modern microscope is a confocal microscope. The spatial resolution is higher than the other types of microscope. The confocal microscope also provides three-dimensional profile of the sample (Cha, Lin, Zhu, Sun, & Fainman, 2000). This technique is highly suitable for getting fresh cell images. The details mechanism and the measuring technique of these devices cannot be provided here because of space limitation. The readers are referred to the literature for details (José Miguel Aguilera, 2005; Perré, 2011).

8.2 X-Ray Tomography

Classical X-ray tomography is based on the attenuation of an X-ray beam by matter which is a coupled effect of photon energy, atomic number, and linear density. An X-ray attenuation system basically consists of a source and a detector. In conventional X-ray tubes, an electron beam, accelerated by a high voltage, collides with a metal target which is known as an anode and produce X-ray photons. In synchrotron facilities, the X-ray beam is produced by the deviation of a high-energy electron beam by intense magnetic fields perpendicular to the beam direction (Falcone et al., 2004). Synchrotron X-ray beams have high brilliance, a high level of polarization, and small angular divergence. Punctual, 1-D, and 2-D sensors have proportional

counters, solid-state detectors, or scintillator coupled to a CCD camera. All these detectors can form 2-D attenuation images of an object, directly or by scanning over one or two spatial dimensions. Three-dimensional description of the object is obtained by computed tomography (CT). A suitable algorithm is used to compute in 3-D (voxels) the radio density values, also called the CT-numbers, from several 2-D projections, grabbed at different rotation angles of the sample (materials science) or of the tube detector set (medical application and materials science). CT scanning was invented independently in the 1970s by Hounsfield and Cormack, who shared the Nobel Prize for medicine in 1979 (Perré, 2011).

Although the first application of computed tomography was primarily in medicine, it rapidly became a very useful tool in physics, materials science and engineering (Wood, Zerhouni, Hoford, Hoffman, & Mitzner, 1995). The initial device has undergone substantial improvement both from technical and computational points of view. Nowadays, the typical resolution of a medical scanner is below one millimeter. This relative accuracy is excellent for imaging objects with a diameter of up to half a meter (Perré & Huber, 2007). However, this resolution is unsuitable for studying porous media. During the last two decades, micro-tomography, based on micro-focus X-ray sources, has become a relatively common tool in materials science (Limodin et al., 2011).

Today, the commercially available desktop equipment allows non-destructive 3-D reconstruction at a spatial resolution of a few mm. A laboratory Nano-CT scanner, which uses an open-type X-ray tube with sub-micrometric spatial resolution, has even been heralded recently. However, in the domain of micro-tomography, the specifications proposed by synchrotron radiation facilities,

which produce monochromatic X-ray beams of high intensity, remain incomparable. At the European Synchrotron Radiation Facility (ESRF), at least two beam lines propose tomography (Baruchel et al., 2006). These two beam lines are ID 19 and ID 15. As the main disadvantage of micro-tomography is its acquisition time, it is worth noting that beam line ID 15 provides a full 3-D data set in less than one second with a spatial resolution. It is also important to mention phase imaging, which allows a much better contrast for light materials such as polymers or cellular plant tissues.

8.3 Nuclear magnetic resonance (NMR)

For determining the portion of water in living cells nuclear magnetic resonance (NMR) is an efficient tool. The physical and chemical environment inside the material affects the relaxation time and the NMR spectra. The details of water in the heterogeneous system can be obtained by NMR (Chatakanonda, Dickinson, & Chinachoti, 2003; Li, Dickinson, & Chinachoti, 1998). The relaxation time of NMR gives useful information regarding the changes in water during food processing.

It has been reported that for the same moisture water is less mobile in gluten than in starch (Li, Dickinson, & Chinachoti, 1996). The NMR technique has been used in determining the water activity in meat. The relaxation time for meat shows multiple exponential decays which means there are different types of water inside the cell (Bertram, Dønstrup, Karlsson, & Andersen, 2002). The researchers have termed the types of water as bound water and strongly bound water in cells (Bodurka, Buntkowsky, Olechnowicz, Gutsze, & Limbach, 1996; Konishi, Kobayashi, & Miura, 2010; Miedziejko, Plenzler, NAPIERAŁA, & NAROŻNA, 1996; Yoshikawa & Ohsaka, 1979).

These two types of water contain different dynamic behavior. For the cheese drying the moisture, profile was determined by the NMR with mass transfer modeling. The proposed model showed good results for both moisture profile and the drying curve (Chen, Wei, & Zhang, 2010).

8.4 NIR Spectroscopy

Near Infrared Reflectance (NIR) spectroscopy operates based on absorption of electromagnetic radiation. NIR spectra of foods comprise combinations of vibrational modes involving numerous chemical bonds (Huang, Yu, Xu, & Ying, 2008). Due to the sensitivity of absorbance in this technology the particle size, chemical state, and moisture content can be tracked effectively. This makes measurements in organic and biological systems feasible using NIR spectroscopy. The schematic diagram of NIR spectroscopy is shown in Figure 13. The radiation may be absorbed, transmitted, or reflected when interacting with samples. Therefore, there are different NIR spectroscopy measurement modes for different applications. NIR spectroscopy is a fast, non-destructive analytical technique and is becoming one of the most promising for online product evaluation (Gou et al., 2013; Stawczyk, Munoz, Collell, & Comaposada, 2009).

NIR spectroscopy is one of the most efficient tools for control and monitoring of food processing system (Huang et al., 2008). The quality of the food products can also be measured by NIR spectroscopy. Continuous development of NIR spectroscopy has made it one of the most robust technologies (Demers, Gosselin, Simard, & Abatzoglou, 2012). During the drying process, this technology can be used to measure the moisture content of the product with the sensitive probes. This approach will be an effective approach for analyzing the active drying phenomena. NIR spectroscopy was used for determining the water activity of sausage during drying (Stawczyk et

al., 2009). The output of the research was impressive, and it proved that NIR spectroscopy is a fast method to control and monitor the drying process. For analyzing the control and monitoring of the microwave drying of Apple thermography assisted NIR spectroscopy was also used (Cuccurullo, Giordano, Albanese, Cinquanta, & Di Matteo, 2012). This system allows an efficient temperature control by detecting the maximum temperature inside the sample.

9. Challenges in Multiscale modeling

Though multiscale modeling has a great potential in explaining the phenomena during drying, it has not been used much in drying technology field (Strumillo, 2006). There are some challenges in implementing the multiscale model. Coupling of the sub-models that describes the related physics in each scale is one of the most important challenges. Specially, building bidirectional relationship between microscale and microscale is a challenge (Perré, 2007). In most of the cases, the coupling has been performed using numerical experiments. For reflecting the actual phenomena boundary condition plays a great role (Nassehi & Parvazinia, 2011). So far dirichlet boundary conditions have been used for this purpose. Researchers have shown that periodic boundary condition exhibit a better result (Yue & Weinan, 2007). On the other hand, Conventional multiscale simulation methods mostly rely on the homogenization process (Pavliotis & Stuart, 2008). The homogenization process separates the scale in term of time and lengths. The size of the representative elementary volume can be a great challenge for establishing the serial coupling (Ingram, Cameron, & Hangos, 2004). Therefore, the representative elementary volume needs to be defined carefully.

The computational domain is an important part of the modeling purpose. An accurate representative elementary volume can represent a better platform for the computation (Quang Tri Ho et al., 2011). There is structural heterogeneity in the microstructure of the food materials. As an example Apple parenchyma cells have a great variation in their size and shape. Therefore, the statistical technique can be used to build the microstructure of the food materials. Moreover, structural heterogeneity consideration is a great challenge in simulating the multiscale model. Researchers have recommended for the finite-element method for the propagation of these random variations (Quang T Ho et al., 2013).

Prediction of the food properties is also a big issue in multiscale modeling. During the drying process the phase transition like boiling or freezing manipulates the material properties. A description of this process is required to simulate the evolution of the phase in microscale (Van der Sman & Van Der Goot, 2009). Recently, researchers have given some description regarding the phase transition during the drying process (Curcio, 2010; Dhall, Squier, et al., 2012; Kumar, Joardder, Farrell, Millar, & Karim, 2014; Kumar et al., 2015; J. Zhang & Datta, 2004). These models are known as multiphase models. The evolution of the microstructure can be presented in multiscale with the help of the multiphase models. Another important factor that can hamper in the path of multiscale modeling is the cell level material properties (M. U. Joardder, Karim, Kumar, & Brown, 2016b). Though, we have a lot of material properties at room temperature, the properties during active drying is still unavailable (Gharsallaoui, Roudaut, Chambin, Voilley, & Saurel, 2007; Mayor & Sereno, 2004; Scheerlinck et al., 2000; Wang & Brennan, 1995). It needs a good number of experiments and a good experimental set up to find out the materials properties

during the active drying process. Lack of proper experimental set up can be a great difficulty to validate the models in active drying condition.

Modeling and numerical simulation of drying of a lot of food materials are much more complicated than modeling and simulation of single food material drying. It needs higher computational cost (Fletcher et al., 2006). Knowledge about multi-phase flow in porous media with phase change phenomena is needed for modeling of drying of food materials (Datta, 2007a). Also, the interaction between particles as well as liquid and gas flows in porous media must be taken into account in the modeling of drying of food materials (Karunasena, Senadeera, Gu, & Brown, 2014). Moreover, the information about the structure and properties of food particles as well as the phenomena occurring at pore scale is needed to provide for better interpretations of food behaviours during drying (José Miguel Aguilera, Chiralt, & Fito, 2003). Because of these complexities, the modeling of the drying process of food materials in a dryer is still challenging. One of the major difficulties related to explaining the water transport phenomena in the intercellular territory. This causes inaccuracy in designing and the validation of the mathematical model and low efficiency of the whole process. A multi-scale model is required to capture all of these intra-particle gradients to describe the processes more accurately.

10. Suggestions for future research

The future of the multiscale modeling approach in food drying sector is promising. It is not expected that the multiscale modeling approach will replace the laboratory or the full-scale testing. The multiscale modeling approach can be very helpful in optimizing the food drying process.

Some studies are found in the field of horticulture and osmotic dehydration that considered only the mass transfer phenomena in a different scale (Aregawi et al 2013, Fanta et al 2014). However, no work has been done in drying area. There is a big research gap in formulating the multiscale model in the field of food drying process where coupled heat and mass transfer should be considered. Moreover, shrinkage is also an important factor during drying. For considering the shrinkage phenomena, it is recommended that the cell mechanics model should be integrated with the heat and mass transfer process. Multiphysics modeling approach and the CFD analysis can be very helpful to develop the multiscale drying model.

The successful implementation of multiscale model will facilitate the energy smart drying process. The required energy to dry the food materials can be easily identified with multiscale modeling technique. Determination of energy required in microscale and macroscale during drying has also a great research prospect in multiscale modelling technique.

11. Conclusions

Multiscale modelling is a new approach in the field of drying technology. In this article, an overview of multiscale modelling in drying technology has been presented. Food materials are considered as porous hygroscopic materials. Therefore, drying of food materials are considered as multiphase and multiscale phenomena. Advanced modelling and simulation techniques are needed to build a multiscale food drying model. One of the most important challenges in formulating multiscale modelling is the cellular level dynamic material property. Another big challenge associated with the multiscale model is its validation. A better experimental set up is required to validate the developed multiscale model. In the other fields, especially in materials

science field, multiscale model has become the mainstream methodology for customizing the process design. The prospect of multiscale model in food drying technology is also highly promising. The multiscale model will be very useful in optimizing the use of current dryer as well as the drying process. It can be a path-way to an environment friendly and energy efficient drying technology. Though multiscale modelling is an emerging modelling technique in the field of food drying technology, its progress can be delayed for the lack of understanding of the physics of food at microscale. Therefore, it is recommended that a detailed research should be continuing in the microscale level and food microstructure.

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Table 1. Water transport mechanism inside food materials during dehydration

Moisture Transfer During Drying		Scale	Reference
Diffusion	Vapor	Macroscale	(Engels, Hendrickx, De Samblanx, De Gryze, & Tobback, 1986; Lewis, 1921)
		Microscale	(Nieto, Castro, & Alzamora, 2001)
	Surface	Macroscale	(Jayasundera, Adhikari, Aldred, & Ghandi, 2009; Zogzas, Maroulis, & Marinos-Kouris, 1994)
	Liquid	Macroscale	(Raisul Islam, Ho, & Mujumdar, 2003)
		Microscale	(Nieto et al., 2001; Shi & Le Maguer, 2002)
Capillary		Macroscale	(Clement K Sankat & Francois Castaigne, 2004)
Evaporation condensation		Macroscale	(Bhandari & Howes, 1999)
Hydraulic flow		Macroscale	(Clement K. Sankat & Francois Castaigne, 2004)
Molecular diffusion		Microscale	(José Miguel Aguilera, Chiralt, & Fito, 2003; Karel & Saguy, 1991)
Knudsen flow		Microscale	(Sagara, 2001)
		Macroscale	(Rossello, Canellas, Simal, & Berna, 1992)
Mutual diffusion		Macroscale	(Sano & Yamamoto, 1993)
Slip flow		Macroscale	(Waananen, Litchfield, & Okos, 1993)

Table 2: Basic physics for single phase model

Physics	Equation
Energy conservation	$\rho c_p \left(\frac{\partial T}{\partial t} + u \cdot \nabla T \right) = \nabla \cdot (k \nabla T) + H$ Where, ρ = density, c_p = specific heat of the material, T = temperature, u = convective flow, K = thermal conductivity of the material, H = heat source, t = time
Mass conservation	$\frac{\partial c}{\partial t} + \nabla \cdot (-D_{eff} \nabla c) + u \cdot \nabla c = R$ Where, c = moisture concentration, D_{eff} = effective moisture diffusivity, R = production or consumption of moisture.

Table 3: The basic governing equation for multiphase modelling

Physics		Governing Equation	Nomenclature
Mass conservation	Liquid water	$\frac{\partial}{\partial t}(\phi S_w \rho_w) + \nabla \cdot (\bar{n}_w) = -R$ $\bar{n}_w = -\rho_w \frac{k_w k_{r,w}}{\mu_w} \nabla P - \rho_w \frac{k_w k_{r,w}}{\mu_w} \frac{\partial p_c}{\partial c_w} - \rho_w \frac{k_w k_{r,w}}{\mu_w} \frac{\partial p_c}{\partial T}$ (a) Convective flux (b) Capillary flux due to concentration gradient (c) Capillary flux due to temperature gradient	S = Saturation, P = Pressure, μ = Dynamic Viscosity, ϕ = Porosity, ρ = Density, k = Permeability D = Diffusivity, R = Evaporation Q = Heat source ω = Mass fraction K = Thermal conductivity, C_p = Specific heat t = time
	Water vapor	$\frac{\partial}{\partial t}(\phi S_g \rho_g \omega_v) + \nabla \cdot \left(-\rho_g \omega_v \frac{k_g k_{r,g}}{\mu_v} \nabla P - \phi S_g \rho_g D_{eff,g} \nabla \omega_v \right) = R$	
	Air	$\omega_a = 1 - \omega_v$	
Energy conservation		$\rho_{eff} c_{peff} \frac{\partial T}{\partial t} + \nabla \cdot (\bar{n}_g h_g + \bar{n}_w h_w) = \nabla \cdot (k_{eff} \nabla T) - h_{fg} R_{evap} + Q$	
Momentum Conservation		$\frac{\partial}{\partial t}(\rho_g \phi S_g) + \nabla \cdot (\bar{n}_g) = R_{evap} , \quad \bar{n}_g = -\rho_g \frac{k_g k_{r,g}}{\mu_i} \nabla P ,$ $\rho_g = \frac{PM_g}{RT}$	<u>Subscript:</u> V = vapor g = gas a = air w = water s = solid eff = effective r = relative

Table 4: Advance modelling approach

Authors	Materials	Application area
Nicolai et al.	Apple	Microstructure characterization
Defraeye et al	Apple	Water transport model
Halder and Datta	Potato slab	Multiphase porous media flow
Chaminda Prasad Karunasena et al.	Apple	Microscale model (2D) for drying
Varabereke et al	Apple	Microscale water transport
Ho et al.	Pear	Microscale gas exchange
Van Ieperdeke	Apple	Calculating mechanical properties of cell
Patrick Perre et al	wood	Multi-scale model in wood drying
Carr et al.	wood	Dual scale model in wood drying

Table 5. Advance non-invasive technique for observing the drying process

Experimental Technique	Spatial resolution	Information Type
MRI	20 μm	Water density, velocity, properties
NIR	10 μm	Moisture content, particle size
NMR	1 μm	Moisture content and the drying rate
X ray micro tomography	0.2 μm	Deformation, crack formation, moisture content, structure of the material
SEM	3nm	Deformation, shrinkage observation
TEM	1A	Deformation, shrinkage observation

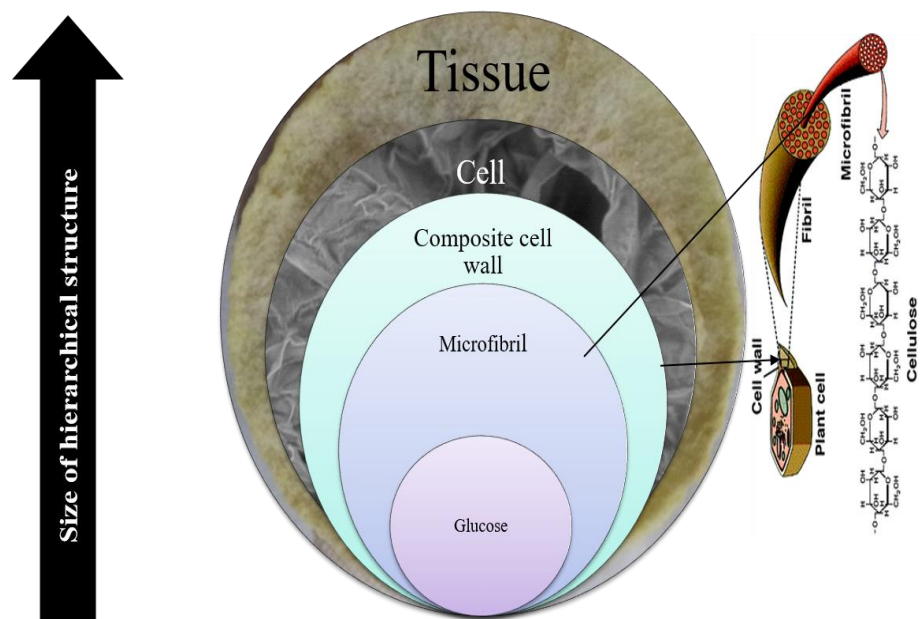


Figure 1. Demonstration of hierarchical structure of cellular based food materials (M. U.

Joardder, Karim, Kumar, & Brown, 2016a)

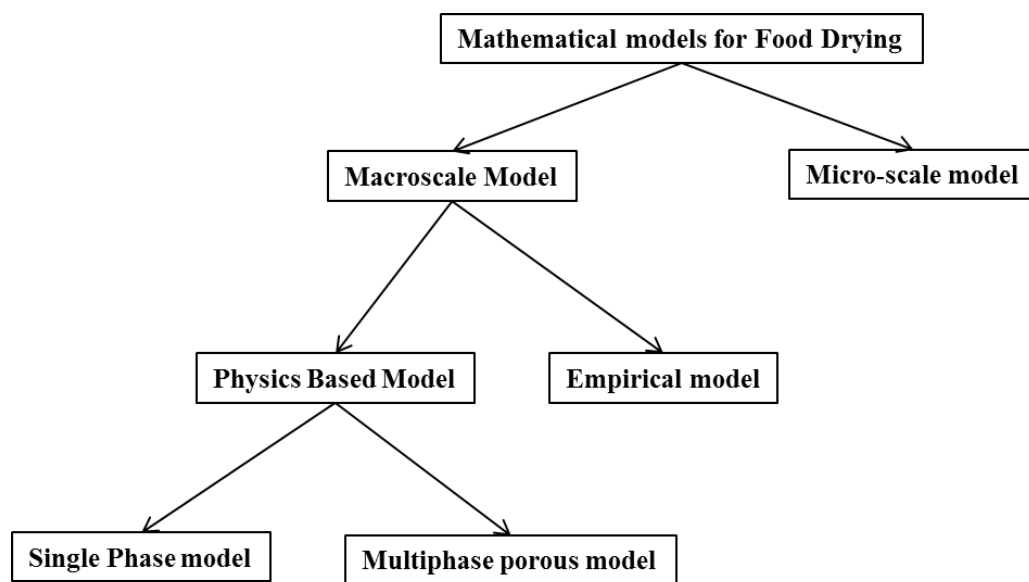


Figure 2. Classification of mathematical models in food drying

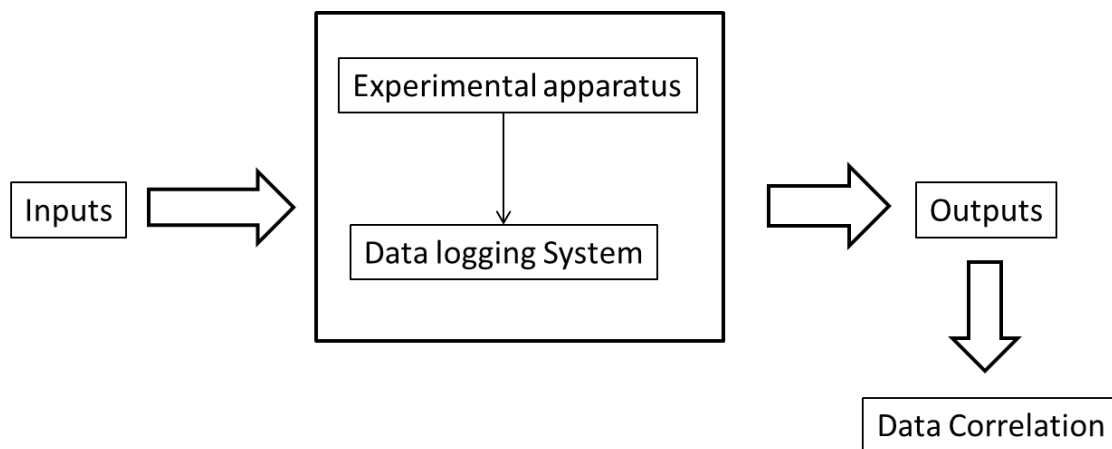


Figure 3. Empirical modeling approach

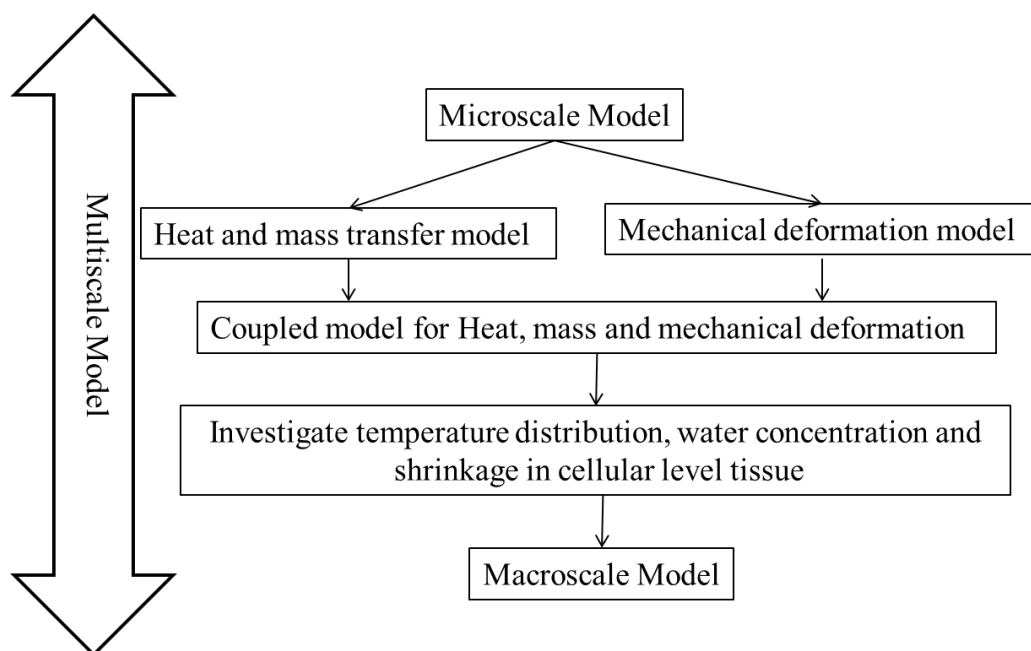


Figure 4. Concept of Multiscale modeling in drying technology

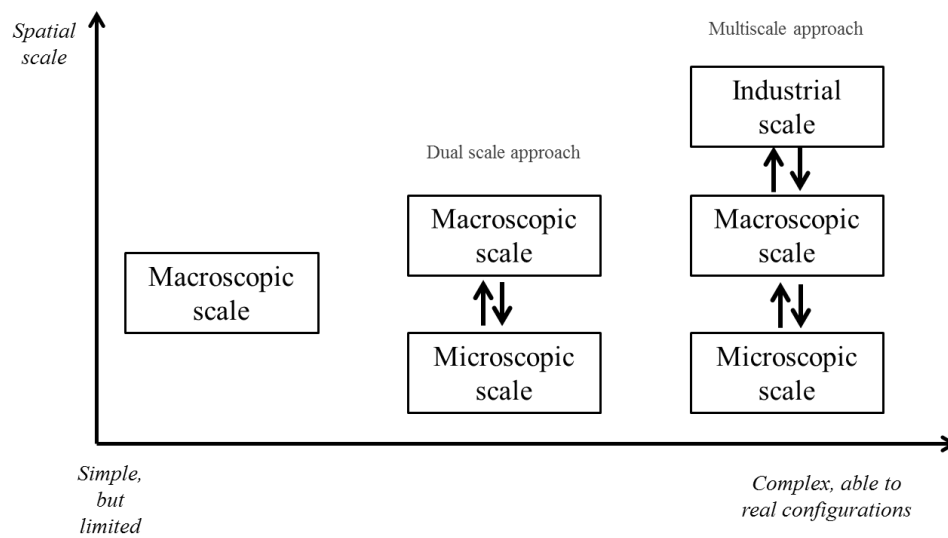


Figure 5. Multi scale approach presented by Patrick Perre (Perré, 2010).

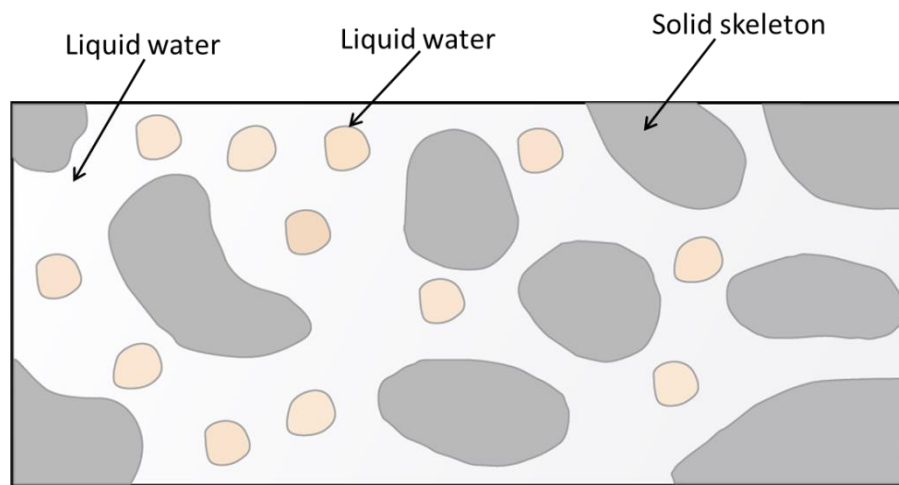


Figure 6: Conventional RVE used by researchers (Datta, 2007a)

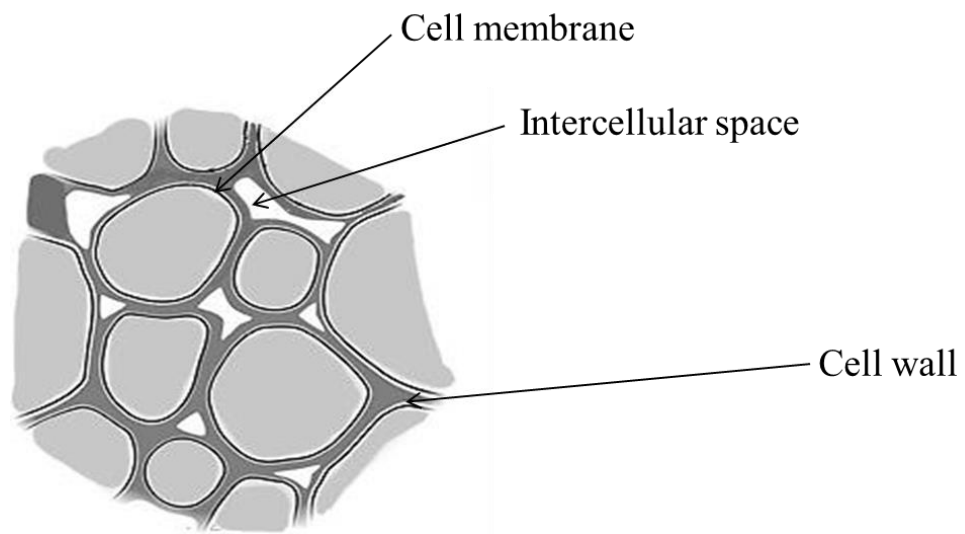


Figure 7. Cellular structure of Plant tissue

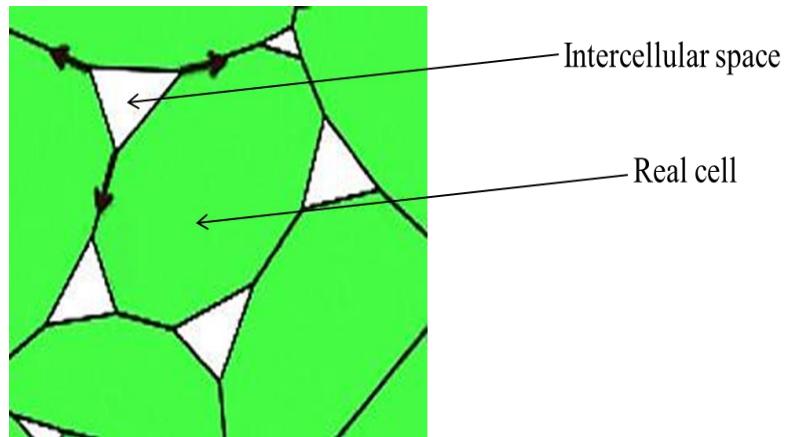


Figure 8. Schematic diagram of actual geometry of the microstructure of plan based tissue

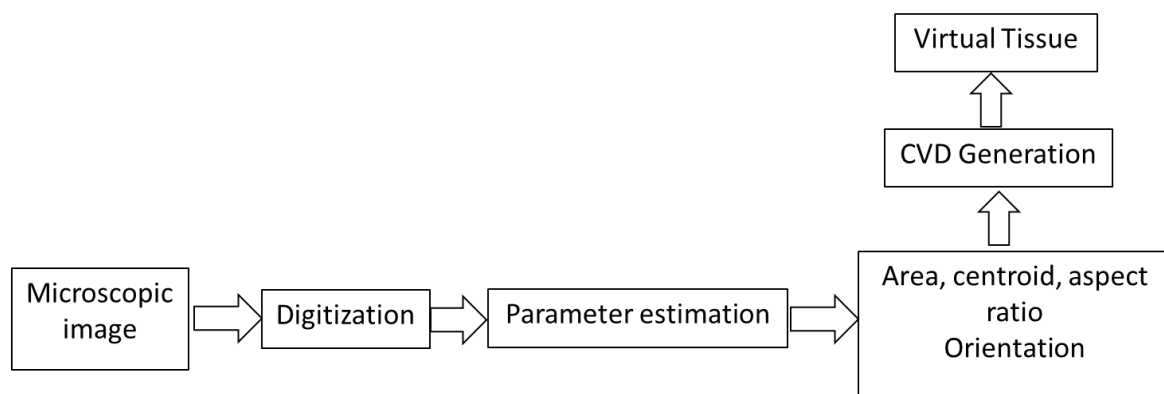
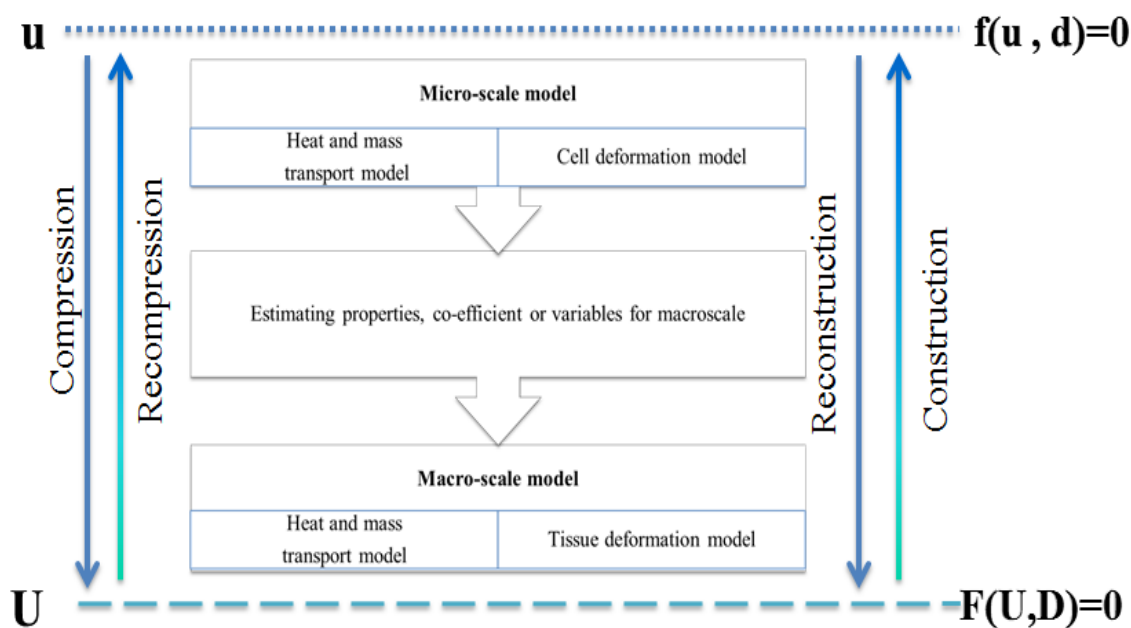


Figure 9. A step by step representation for modeling of cell geometry



10: Heterogeneous multiscale method framework

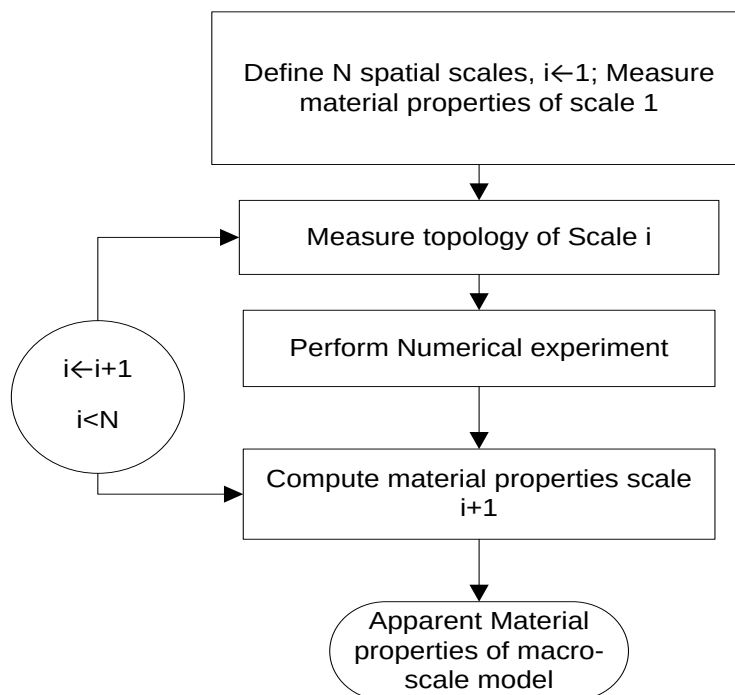


Figure 11. A sample of homogenization process

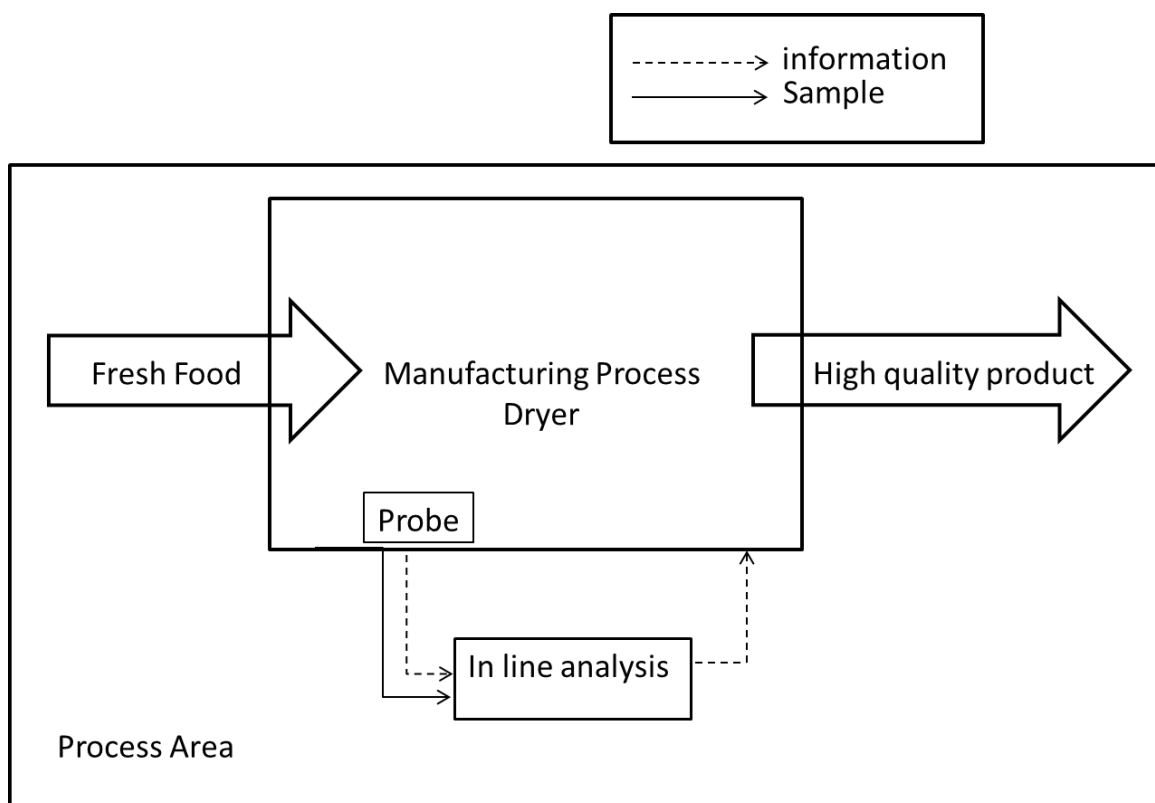


Figure 12: Schematic representation of modern drying system

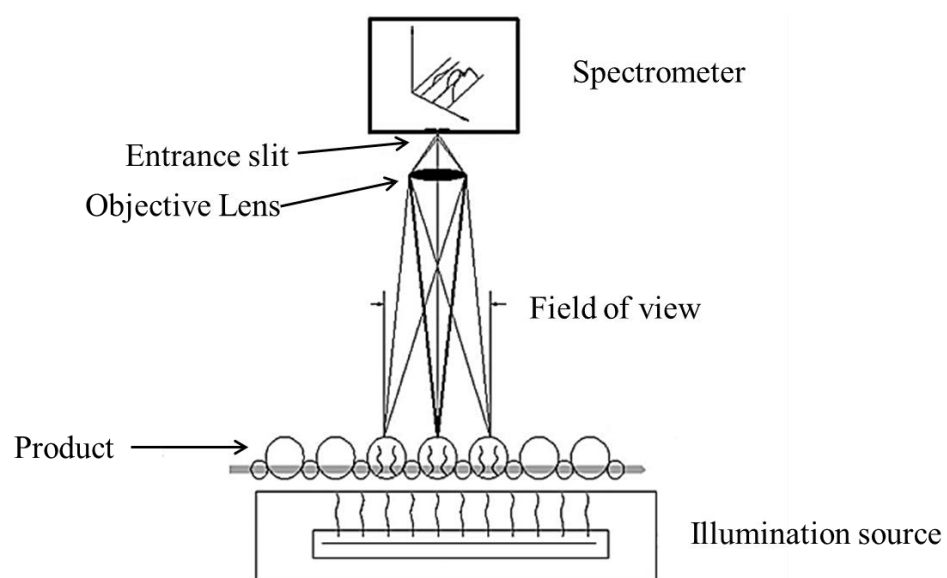


Figure 13. The schematic diagram of NIR spectroscopy